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1 The mathematics of collar neighborhoods

1.1 Basics about collar neighbourhoods

Definition 1.1.1. A *topological manifold* is a second-countable Hausdorff space that is locally Euclidean.

Remark 1.1.2. Apparently, there are some subtleties with collars of non-metrizable manifolds but I assume we don't care. (At least with the above definition our manifolds are always metrizable.)

Definition 1.1.3. Let M and N be two C^k -manifolds. Let $f: M \rightarrow N$.

- (i) f is a C^k -diffeomorphism if it is bijective and both f and its inverse functions are k -times continuously differentiable.
- (ii) f is a C^k -embedding if it is a C^k -diffeomorphism onto its image.

Definition 1.1.4 (Due to [Con71]). Let M be a C^k -manifold (with boundary) and let $B \subseteq M$ be closed.

- (i) B is *locally collared* if it is covered by sets $U \subseteq B$ which are open in B such that there are C^k -embeddings $\varphi_U: \overline{U} \times [0, 1] \rightarrow M$ with
 - a) $\varphi_U^{-1}(B) = \overline{U} \times \{0\}$.
 - b) For all $x \in \overline{U}$, $\varphi_U(x, 0) = x$.
 - c) $\varphi_U(\overline{U} \times [0, 1])$ is closed.
 - d) $\varphi_U(U \times [0, 1])$ is open.
- (ii) B is *collared* if B already fulfils the conditions given above. In this case we call the image of $B \times [0, 1]$ under the corresponding map $\varphi_B: B \times [0, 1] \rightarrow M$ a *collar neighbourhood* of B .

Remark 1.1.5. This definition is a lot more restrictive than other definition which don't consider the closure or the closed interval. I hope this still works.

Remark 1.1.6. For $k = 0$ we can even extend this definition to topological spaces instead of just defining it for topological manifolds.

Theorem 1.1.7 (Due to [Con71]). Let X be a Hausdorff space and $B \subseteq X$ be a closed subset. If B is locally collared in X , then it is collared.

Proof. Let $U_1, \dots, U_n$ be a cover of B that witnesses that B is locally collared. Let $\varphi_i: \overline{U_i} \times [0, 1] \rightarrow X$ be the corresponding embeddings. B is normal because it is compact and Hausdorff. By the shrinking lemma (already in mathlib) there is a new open cover $V_1, \dots, V_n$ of B such that $\overline{V_i} \subseteq U_i$ for all $1 \leq i \leq n$. Define $X^+ := X \amalg B \times [-1, 0]$... $\square$

We would like the following theorem:

Theorem 1.1.8. *Let M be a C^k -manifold (with boundary) and $B \subseteq X$ a closed subset. If B is locally collared in M , then it is collared.*

Then we will need the following lemma:

Lemma 1.1.9. *Let M be an n -dimensional C^k -manifold with boundary. Then ∂M is locally collared.*

Proof. Let $p \in \partial M$. Then p is contained in a regular half-coordinate ball U , i.e. there is a coordinate half-ball $U' \supseteq \bar{U}$ and a coordinate map $\psi: U' \rightarrow \mathbb{H}^n$ such that for some $r' > r > 0$ we have $\psi(U) = B_r(0) \cap \mathbb{H}^n$, $\psi(\bar{U}) = \overline{B_r(0)} \cap \mathbb{H}^n$ and $\psi(U') = B_{r'}(0) \cap \mathbb{H}^n$.

Now pick an open neighbourhood V of $0 \in \partial \mathbb{H}^n$ and $l > 0$ s.t. $V \times [0, l] \subseteq B_r(0)$.

Considering $\bar{V}$ and V as submanifolds of $\mathbb{H}^n$ and $\psi^{-1}(\bar{V}) = \overline{\psi^{-1}(V)}$ and $\psi^{-1}(V)$ as submanifolds. Then

$$\psi|_{\overline{\psi^{-1}(V)}} : \overline{\psi^{-1}(V)} \rightarrow \bar{V}$$

and

$$\psi|_{\psi^{-1}(V)} : \psi^{-1}(V) \rightarrow V$$

are C^k -diffeomorphisms. So it suffices to show that $\psi^{-1}|_{\bar{V} \times [0, l]} : \bar{V} \times [0, l] \rightarrow M$ fulfils the conditions given in Definition 1.1.4.

I think this should work... I think it would be nicer to do this directly in Lean and then later write up the maths proof. $\square$

which will then allow us to derive

Theorem 1.1.10 (Collar neighbourhood theorem). *Let M be a C^k -manifold with boundary, then ∂M is collared.*

Bibliography

- [Con71] Robert Connelly. “A New Proof of Brown’s Collaring Theorem”. In: *Proceedings of the American Mathematical Society* 27.1 (1971), pp. 180–182. ISSN: 00029939, 10886826. URL: <http://www.jstor.org/stable/2037284> (visited on 04/29/2026).